

Dominic Bankovitch

dbankovitch@berkeley.edu | dominicbankovitch.com | linkedin.com/in/dominic-bankovitch/ | github.com/dominic738

EDUCATION

University of California, Berkeley

Berkeley, CA

Major: Data Science; Minor: Computer Science; GPA: 4.00/4.00; Honors to Date Fall 2025

2024 – 2027

- **Relevant Coursework:** Discrete Math, Structure and Interpretation of Computer Programs, Data Structures and Algorithms, Foundations of Data Science, Principles and Techniques of Data Science, Data Engineering, Probability for Data Science, Data Mining and Analytics, Data and Justice

University of Notre Dame

Notre Dame, IN

Applied Mathematics & Statistics; Science Honors Program; GPA: 3.96/4.00; 2x Dean's List

2023 – 2024

- **Relevant Coursework:** Microeconomics, Calculus III, Applied Linear Algebra, Differential Equations

PROFESSIONAL EXPERIENCE

Oracle

Fall 2026

Data Consultant

Berkeley, CA

- Architected data pipeline on **Oracle Cloud Infrastructure (OCI)**, achieving sub-second sensor-to-database latency.
- Built a stream time-alignment system reconciling inertial and video data onto a common timeline despite independent capture clocks and network jitter.
- Ran a fusion ablation study quantifying the accuracy gain from combining camera and sensor data over either alone.
- Powered a live push-up and jumping-jack demo, translating raw sensor data into real-time rep counts and form feedback.

Expedia Group

Summer 2026

Data Science Intern

Seattle, WA

- Quantified traveler rebooking behavior after false fraud declines by building SQL cohorts in **Querybook**.
- Designed a quasi-experiment using **Propensity Score Matching**, **Difference-in-Differences**, and **Meta Learners** to isolate the effect of login security measures on falsely blocked supplier accounts.
- Implemented the causal analysis in **Databricks**, matching accounts on pre-treatment behavior to estimate treatment effect.
- Wrote complex **SQL** joins and long-to-wide transformations in **Trino** to build unified traveler-supplier feature tables from disparate fraud and login datasets.

Kohl's

Summer 2025

Data Science Intern

San Francisco, CA

- Imputed e-commerce attributes across **6M+** products and processed **22M+** SKUs with GPU-parallelized pipelines.
- Built XGBoost gender classifier achieving **0.98 AUC**; extended to multi-label prediction achieving **0.87 F1** across 6 classes.
- Trained Bayesian-optimized Keras neural network on SBERT hierarchy embeddings to classify **2,018** e-commerce paths.
- Achieved **96% Top-5 accuracy** on hierarchy prediction; deployed results to BigQuery via scalable batch pipeline.

Algorhythm

Fall 2025

Machine Learning Engineer

Berkeley, CA

- Encoded video frames with **CLIP** and used a pretrained **MLP** to project them into a 13-dim emotion space.
- Matched each frame to a candidate track via cosine-similarity **nearest-neighbor search** over EMID's emotion vectors.
- Developed **Flask** web app for video upload and real-time music overlay generation, delivered as a 5-person team.
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Deloitte

Spring 2024

Student International Business Council Consulting Extern

Notre Dame, IN

- Organized AV market simulation, researching adoption strategy, sensor tech, and competitors - presented to Deloitte.

SELECTED PROJECTS

The Musical Compass | FastAPI, OpenAI, scikit-learn, Pinecone

Present

- Generated custom semantic axes on demand by training an **SVM** on GPT-4o-mini-generated phrase embeddings.
- Embedded lyrics with **OpenAI** embeddings, weighting repeated lines (hooks, choruses) more heavily per song.
- Cached embeddings in **Pinecone** to avoid redundant LLM calls, saving **5K tokens** and **30s** per repeat axis lookup.
- Integrated Spotify API with regex cleaning; deployed **FastAPI** backend on Google Cloud Run, frontend on Netlify.

Album Mosaic Filter | Python, KD-Tree, Computer Vision

Present

- Built real-time video-to-mosaic generator over 5,000+ album covers, mapping each pixel to its perceptually closest cover.
- Indexed album artwork using a **KD-tree** in **LAB color space** for efficient nearest-neighbor color matching at pixel scale.

TECHNICAL SKILLS

Languages: Python, SQL, Java, TypeScript, JavaScript, HTML/CSS

Libraries/Frameworks: pandas, NumPy, scikit-learn, XGBoost, LightGBM, CatBoost, PyTorch, TensorFlow, Keras, Flask, FastAPI, React, Tailwind CSS, OpenCV, PySpark

ML/AI: Classification, Neural Nets, NLP (Transformers, SBERT), CLIP, OpenAI API, Feature Engineering, A/B Testing, RAG

Tools/Databases: Git, Jupyter, GCP, Vertex AI, MySQL, BigQuery, Redis, Databricks, Pinecone, Docker, Tableau